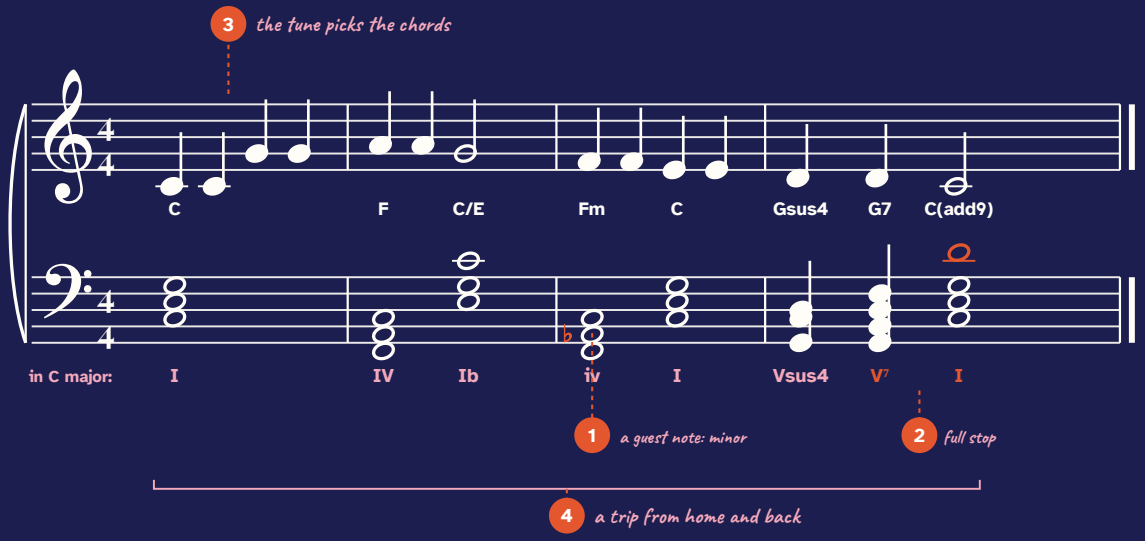
PART V · PUT THEM TO WORK

# Harmony in music

A melody is the singer; harmony is the band behind it. Here the chords start moving: guests from outside the key, endings, chords under a tune, and the trips every song takes. All of it is in the same four bars.





1 **Chords from outside the key** p. 113
a guest note: A♭ in C major

2 **Cadences** p. 116
V to I ends a phrase like a full stop

3 **Chords under a melody** p. 119
the tune's main notes pick the chord

4 **Progressions everywhere** p. 121
the same few trips, in every song

★ **Tips: when harmony goes wrong** p. 123

PART V · HARMONY IN MUSIC · 1

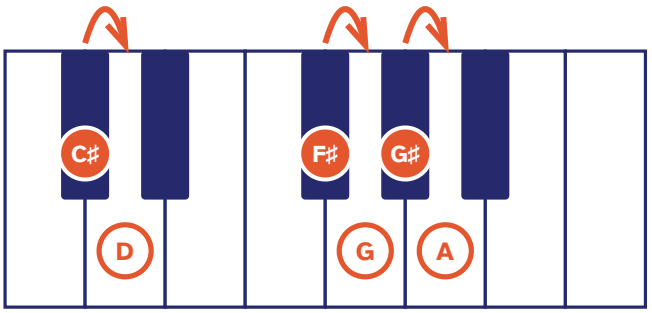
# Chords from outside the key

In C major, home is the white keys, so a black key in a chord is a **guest**. Guests never stay: each one moves to the **very next key** (no key between), and shows you where the music goes next.

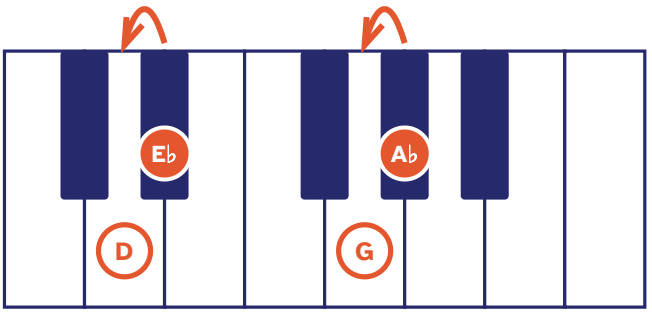
**THE SECRET**

**Sharp guests push up, flat guests sigh down: follow them one key and you find the next chord.**

### Sharp guests push up

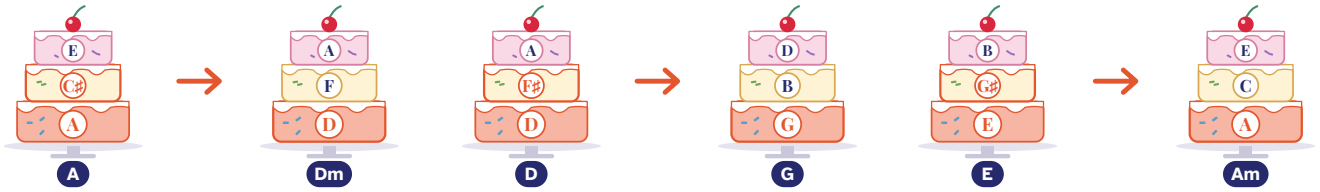



### Flat guests sigh down

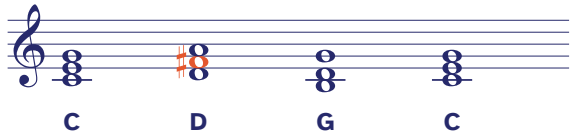



1. Find the **black key** in the chord: that is the guest.
2. A **sharp**? Go **up** to the very next key. A **flat**? Usually **down** to the very next key.
3. You have found a note of the **next chord**. Play it and hear it land.

## Sharp guests: up one key names the next chord

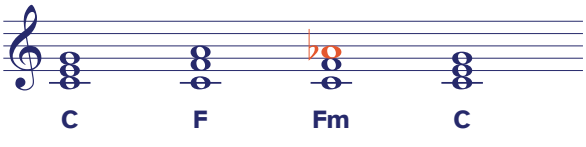



C♯ goes up to **D**, so the next chord is D minor. F♯ goes up to **G**: G. G♯ goes up to **A**: A minor.




Try it in a song: C, D, G, C. The **F♯** pushes up into G, then G leads home to C.

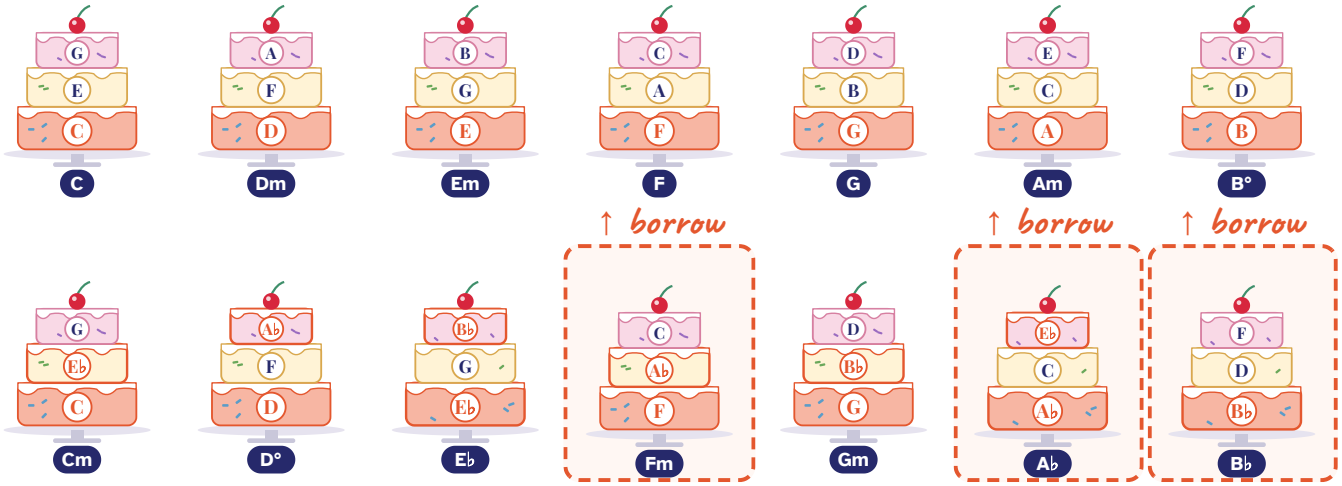
## Flat guests: borrowed from the twin, they sigh down




C major and C minor are **twins** (the parallel minor): the same home note, C, in different clothes. A flat guest is one piece of the twin's outfit. In C, F, Fm, C the **A♭** sighs down to G.

PART V · HARMONY IN MUSIC · 1 · CONTINUED

## C major *at home*




## C minor *the twin: same home note, darker clothes*

## Their names, for later

| GUEST | NOTES | MOVES | NEXT CHORD | ITS NAME |
|---|---|---|---|---|
| **sharp guest** | D F♯ A | F♯ up to G | G | V of V |
| **sharp guest** | A C♯ E | C♯ up to D | Dm | V of ii |
| **sharp guest** | E G♯ B | G♯ up to A | Am | V of vi |
| **flat guest** | F A♭ C | A♭ down to G | C | borrowed iv |
| **flat guest** | A♭ C E♭ | A♭ down to G | G or C | borrowed ♭VI |
| **flat guest** | B♭ D F | B♭ on to C | C | borrowed ♭VII |

B♭ is the flat guest that walks **up** instead, on to C (one key between: a tone): the rock and film ending B♭ → C.

EXAMPLE

### Where do the guests go? B D♯ F♯ in C major



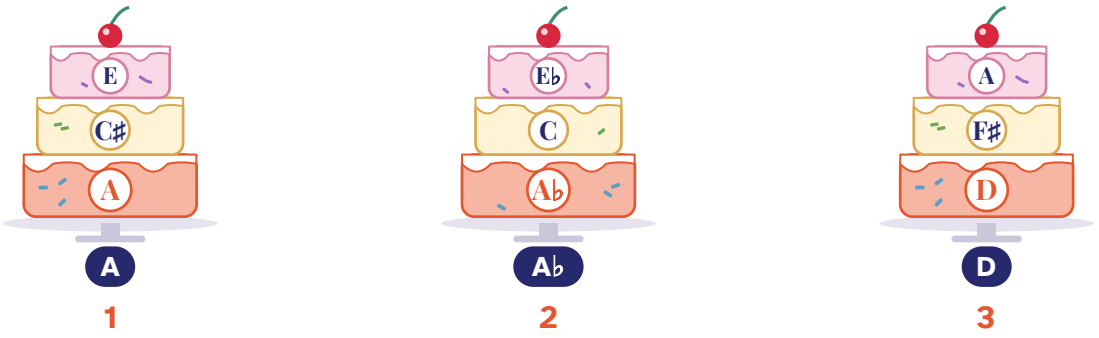

1. **Find the black keys:** there are two guests here.
2. **Both are sharps:** move each up to the very next key. Where do they land?
3. **Which cake in C major** holds both landing notes?

D♯ goes up to E, F♯ up to G: both live in **E minor** (E G B). B D♯ F♯ leads to Em.

PART V · HARMONY IN MUSIC · 1 · CONTINUED

GAME

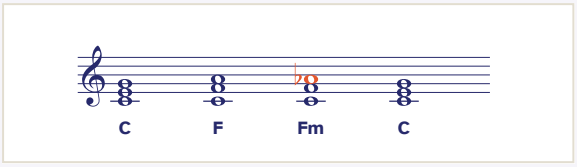
## Where does the guest go?




In C major: find each guest, follow it one key, and write the chord that comes next.

Answer: 1 C♯ up to D: D minor · 2 A♭ down to G, E♭ down to D: G · 3 F♯ up to G: G.

TRY IT · 10 SECONDS




### Play C, F, Fm, C slowly.

Hear the A♭ sigh down to G? Now play C, D, G, C: the F♯ pushes up instead. One sighs, one pushes.

THINK OF IT A surprise guest at dinner: one new face changes the whole conversation, and then everyone goes home.

TRAP

- × Following a sharp downwards: sharps push **up**, flats usually sigh **down**.
- × Calling every black key a guest: in G major F♯ is at home. A guest is a note from outside the key you are in.
- × Learning the names first: play it, hear where the guest goes, then name it.
- × Mixing up the relatives: C minor is C major's twin (same home note). A minor is its neighbour (same key signature). We borrow from the twin.

SEE ALSO

Chords in a key p. 93 · Slash chords p. 108 · Cadences p. 116

PRACTISE IT

Workbook Harmony 9 · Online: Library › Chromatic chords

PART V · HARMONY IN MUSIC · 2

# Cadences

A phrase is a musical sentence, and it needs punctuation. The last two chords of a phrase are its **cadence**. They tell you: finished, taking a breath, or surprise!

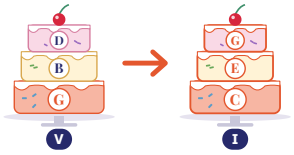
THE SECRET

**Music has full stops and commas too: listen to the last two chords of a phrase.**

.

**Perfect**

V → I

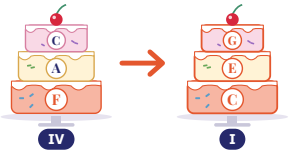



*Home. The end.*

Amen

**Plagal**

IV → I

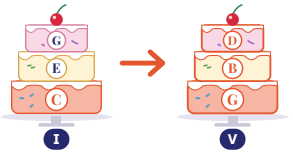



*A soft, gentle end.*

,

**Imperfect**

… → V

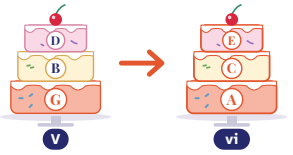



*To be continued…*

?!

**Interrupted**

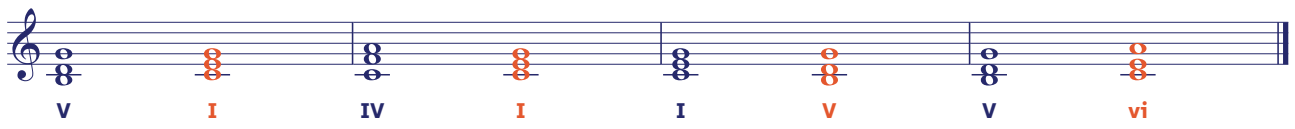
V → vi




*Surprise! Not home.*

1. Find the end of a phrase: often a long note, or a breath.
2. Name its **last two chords** with numerals: V I, IV I, anything then V, or V vi.
3. Match the punctuation: full stop ( . ), Amen, comma ( , ) or ?!

## The four cadences in C

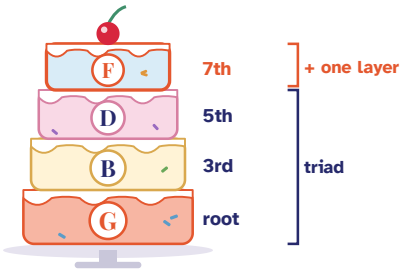


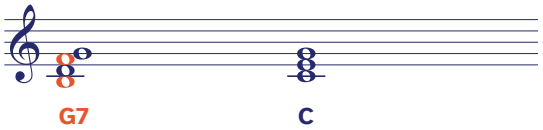

Perfect, plagal, imperfect, interrupted. The red chord is where each one lands.

| CADENCE | CHORDS | FEELS LIKE | ALSO CALLED |
|---|---|---|---|
| **Perfect** | V → I | a full stop: finished | authentic, full close |
| **Plagal** | IV → I | the “A-men” sung at the end of a song in church | Amen cadence |
| **Imperfect** | any → V | a comma: waiting | half close, half cadence |
| **Interrupted** | V → vi | a surprise: not home yet | deceptive |

PART V · HARMONY IN MUSIC · 2 · CONTINUED

## Why G7 pulls home



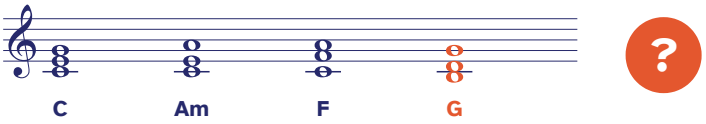

Add the 7th to V and the full stop gets stronger.

In **G7** two notes lean towards C major. The **B** rises to **C**, the **F** falls to **E**: each to the very next key, no key between. Two tiny steps, and you are home (Seventh chords, p. 99).

It works like a guest: B pushes up, F sighs down (p. 113).

EXAMPLE 1

### How does this phrase end? C Am F G




1. **Find the last two chords** of the phrase.
2. **Turn them into numerals** in C major.
3. **Which punctuation card** ends that way?

F then G is IV → V: it lands on V, the comma. An **imperfect cadence**.

EXAMPLE 2

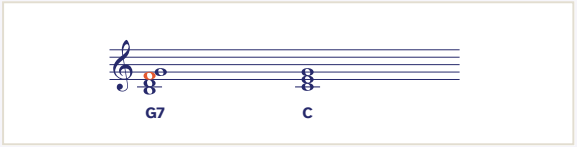
### How does Twinkle end?




1. **The last two chords** of Twinkle in C are G, then C.
2. **Turn them into numerals.**
3. **Which card** goes from that chord to that chord?

V → I: a **perfect cadence**, the full stop. That is why Twinkle sounds finished.

TRY IT · 10 SECONDS




### Play G7, then C. The end!

Now play G7, then A minor: “huh?!” That is the interrupted cadence. Then play F, then C, and sing “A-men”.

PART V · HARMONY IN MUSIC · 2 · CONTINUED

EXAMPLE 3

### Surprise! G7, then A minor

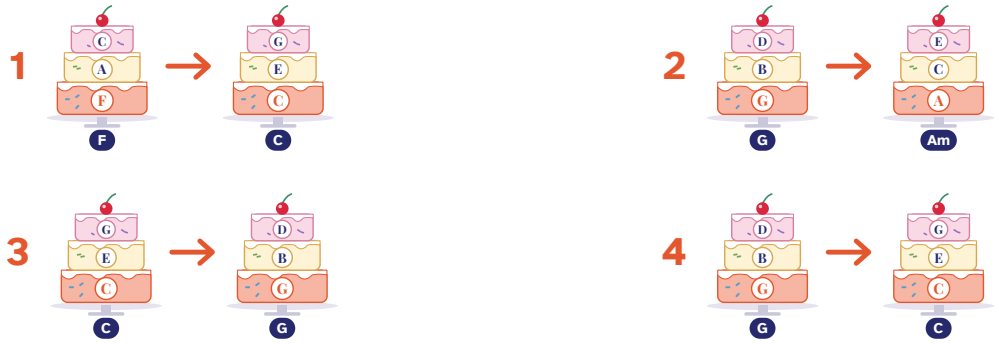



1. **G7 leans towards home**: you expect C.
2. **But the next chord is A minor.** Which numeral is it in C?
3. **V → that numeral:** which card?

V → vi: an **interrupted cadence**. The music was heading home, then turned the corner.

GAME

### Punctuate the phrases




Each pair ends a phrase in C major. Turn the letters into numerals first (C is I, F is IV, G is V, Am is vi), then write full stop, comma, Amen or ?! beside each.

Answer: 1 Amen (plagal), 2 ?! (interrupted), 3 comma (imperfect), 4 full stop (perfect).

THINK OF IT Punctuation in a sentence: a full stop ends it, a comma makes you wait, and a surprise makes you gasp. Music ends its phrases the same way.

TRAP

- × Calling every ending on I (C) perfect: a perfect cadence comes **from V**. F to C is plagal.
- × Mixing up imperfect and perfect: imperfect **ends on** V; perfect **starts on** V and lands on I.

SEE ALSO

Chords in a key p. 93 · Seventh chords p. 99 · Chords under a melody p. 119

PRACTISE IT

Workbook Harmony 10 · Online: Library › Cadences

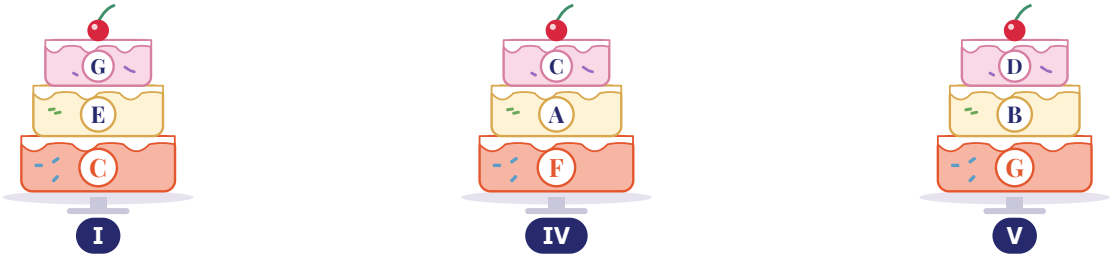
PART V · HARMONY IN MUSIC · 3

# Chords under a melody

The melody is the singer; the chords are the band. To choose a chord for each bar, look at the melody's main notes and pick the cake holding them.

**THE SECRET**

**Look at the melody's main notes, then pick the cake that holds them.**




| C | D | E | F | G | A | B |
|---|---|---|---|---|---|---|
| I, IV | V | I | IV | I, V | IV | V |

*Which cake holds the tune's main notes?*

Every note of C major lives in I, IV or V. C and G live in two: let your ear choose.
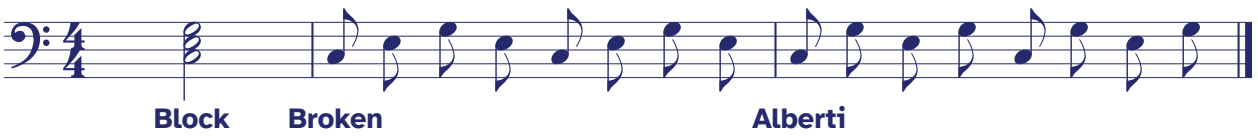
1. Look at the **main notes** of the bar: the long ones, and the ones on beats 1 and 3.
2. Find the cake that holds them: **I, IV or V**.
3. Play that chord in the left hand while the right hand plays the tune.

## Twinkle, Twinkle, with its chords




C C G G sit in I (C); A A in IV (F), then G in I; F F in IV, E E in I; D D in V (G), and the last C in I. The chord changes halfway through bars 2, 3 and 4.

## Three ways to play the chord in the left hand




Block: all three at once. Broken: one at a time, up and down. Alberti: low, high, middle, high, named after Domenico Alberti, who loved it.

PART V · HARMONY IN MUSIC · 3 · CONTINUED

EXAMPLE

**Choose the chord: E D C D | E E E**

E D C D | E E E

red = beats 1 and 3

1. **Bar 1:** which notes fall on beats 1 and 3 (red)?
2. **Which cake** holds both of them: I, IV or V?
3. **Bar 2:** E E E. Which cake holds E?

E and C both live in **I**, and so does E: both bars take **I** (C). That is the start of Mary Had a Little Lamb.

GAME

**Pick the cake**
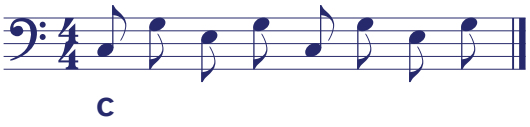
F A F A  D B G B  G E C E

1 2 3

Each bar is in C major. Which cake holds its notes: I, IV or V?

Answer: 1 IV (F A C) · 2 V (G B D) · 3 I (C E G).

TRY IT · 10 SECONDS




**Left hand: C G E G, over and over.**

That is the Alberti bass on the C chord. Keep it going while you hum Twinkle, then try it on F (F C A C) and G (G D B D).

YOU KNOW THIS

**Mozart, Sonata in C, K. 545: the left hand**




The opening left hand is an **Alberti bass**: C G E G is just the C chord, broken into low, high, middle, high; D G F G is G7 (G B D F, without the B). Mozart wrote it an octave higher, starting on middle C.

**Try it.** Play this left hand alone, then add the famous tune: C, E G, B, C D C.

THINK OF IT Like picking a coat for the weather: when the tune changes, change the chord.

TRAP

- × Choosing a chord for a passing note: use the main notes (strong beats, long notes).
- × Playing the left hand louder than the tune: the melody is the star, the chords are the stage.

SEE ALSO

Chords in a key p. 93 · Cadences p. 116

PRACTISE IT

Workbook Harmony 11 · Online: Library › Chords

PART V · HARMONY IN MUSIC · 4

# Progressions everywhere

A **progression** is a trip from chord to chord. A handful of trips turn up in song after song. Learn them by their numbers and you can play them in any key.

**THE SECRET**

**Most songs take the same few trips: home, away, a pull, and home again.**

I home · vi iii home's cousins · IV ii away · V the pull back home

**The pop four**
countless pop songs and ballads

I C → V G → vi Am → IV F

**The fifties**
old rock 'n' roll and doo-wop

I C → vi Am → IV F → V G

**The jazz three-step**
jazz standards (as ii7, V7, Imaj7)

ii Dm → V G → I C

**The canon**
Pachelbel's Canon and its many children

I C → V G → vi Am → iii Em → IV F → I C → IV F → V G

1. Learn the trip by **numbers**: I V vi IV.
2. Pick a key and turn the numbers into chords: in C, **C G Am F**.
3. Play it round and round. Change key: the numbers stay the same.

## Same trip, any key

| The pop four | I | V | vi | IV |
|---|---|---|---|---|
| **in C** | C | G | Am | F |
| **in G** | G | D | Em | C |
| **in F** | F | C | Dm | B♭ |
| **in D** | D | A | Bm | G |

## The twelve-bar blues

| | | | |
|---|---|---|---|
| I C | I C | I C | I C |
| IV F | IV F | I C | I C |
| V G | IV F | I C | V G |

The most famous trip of all is twelve bars long and uses only **I, IV and V**. Almost every blues, and a lot of rock 'n' roll, follows it. Play it in C with one chord per bar, four beats each.

Make it bluesy: turn every chord into a seventh (C7, F7, G7, p. 99).

PART V · HARMONY IN MUSIC · 4 · CONTINUED

EXAMPLE

### Play the pop four in A major

I ? → V ? → vi ? → IV ?

1. **Write the A major scale** (three sharps) and number the notes.
2. **Find notes 1, 5, 6 and 4.**
3. **Build each cake:** which one is minor?

A, E, F♯m, D: **the pop four in A.** Same trip, new key.

GAME

### Name the trip

1 C · Am · F · G

2 Dm · G · C

3 C · G · Am · F

All in C major. Turn each into numbers, then name the trip: the pop four, the fifties or the jazz three-step.

Answer: 1 the fifties (I vi IV V) · 2 the jazz three-step (ii V I) · 3 the pop four (I V vi IV).

TRY IT · 10 SECONDS

I C → V G → vi Am → IV F

### Play C, G, Am, F, four beats each.

Go round four times without stopping, left hand plays the chord's name note (C for C, A for Am), right hand the whole cake. Hum any tune you like over it.

TRAP

× Mixing up the pop four and the fifties: same four chords, different order. Listen to which chord comes second.

SEE ALSO

Chords in a key p. 93 · Cadences p. 116 · Chords under a melody p. 119

PRACTISE IT

Workbook Harmony 12 · Online: Library › Chords

PART V · HARMONY IN MUSIC · TIPS

# When the harmony goes wrong

Every chord can be right and the music still sound wrong: the ending will not close, or the left hand fights the tune. The slip is in how the chords move. Find which move it is, then fix that one.

TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| The end does not feel finished | **the cadence**: end with V → I | p. 116 |
| A sharp or flat surprises you | **a guest**: sharps go up, flats go down, one key | p. 113 |
| The left hand fights the tune | **the main notes**: pick the cake holding them | p. 119 |
| A new song feels hard to learn | **the trip**: find I, IV, V and vi by number | p. 121 |

## Four ways to hear the harmony

**1 Hum the bass**
Sing the lowest notes: they show where the chords move.

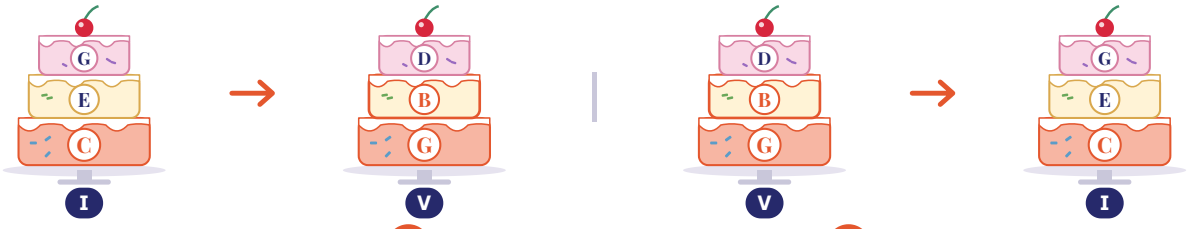
**2 Tick the full stops**
Mark every V → I in your piece with a red tick.

**3 One trip, three keys**
Play I–V–vi–IV in C, G and F: same numbers, new letters.

**4 Chord a tune you know**
Pick I, IV or V for each bar of Twinkle or Happy Birthday.

EXAMPLE

### Fix the ending




1 **The piece stops, but it does not feel finished.** Look at the last two chords.

2 **Name them:** C then G is I → V. That is a comma, not a full stop.

3 **Swap the order:** G then C is V → I. The red B climbs one key to C, and the music lands.

A strong ending needs **V → I** (IV → I for a soft Amen): the last two chords are the full stop.

THINK OF IT Like a road trip: once you know where home is, every detour makes sense, and every road leads back to I.